

Adham Haythem

Mechatronics Engineering Student

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🌐 <https://adham-elmessiry.github.io/Portofolio/>

Professional Summary

Mechatronics Engineering student with practical experience in robotics, embedded systems, artificial intelligence, automation, automotive diagnostics, and intelligent control systems. Experienced in integrating software (Python, C++, MATLAB, OpenCV) with mechanical and electronic hardware to develop real-world engineering solutions. Passionate about intelligent transportation, industrial automation, AI-driven robotics, and smart infrastructure technologies.

Education

Canadian University Dubai (CUD)

Dubai, UAE

Bachelor of Engineering - Mechatronics Engineering

Expected Graduation: 2027

- Relevant coursework: Intelligent Systems, Mechanical Vibration, Control Systems, Embedded Systems, Robotics, Engineering Economy, Signal Processing.

Professional Experience

AGMC BMW - Engineering Intern

Dubai, UAE

- Performed advanced vehicle diagnostics using BMW ISTA, including ECU fault tracing and CAN Bus analysis.
- Assisted senior technicians with mechanical and electrical troubleshooting, vehicle calibration, and repair procedures.
- Developed practical knowledge of automotive control systems, electronic modules, and vehicle communication networks.

Founder - Maktoub Clothing Brand

Dubai, UAE

- Founded and developed an independent clothing brand, managing product design, branding strategy, marketing, and digital presence.
- Applied entrepreneurship principles including customer analysis, brand identity development, and social media marketing.

Competitions

Roads and Transport Authority (RTA) Innovation Competition

2026

- Collaborated in a multidisciplinary team to solve a real-world transportation challenge during a one-week innovation competition.
- Designed an AI-powered intelligent traffic management solution focused on reducing congestion and improving emergency vehicle response.
- Developed a live proof-of-concept demonstrating adaptive traffic signal optimization using AI technologies including Mistral AI.
- Contributed to the Business Plan, Marketing Strategy, Responsible AI framework, and Data Management strategy presented to RTA judges.

Technical Projects

AI Object Detection System

- Developed and trained an object detection model using a custom dataset of approximately 800 labeled images over 50 epochs.
- Built a web-based interface to test the trained model and visualize detection results.

AI Autonomous Robocar

- Developed a Raspberry Pi-based autonomous vehicle integrating camera vision and ultrasonic sensors.
- Implemented Python-based navigation and lane detection algorithms for autonomous movement.

Intelligent Automated Package Sorting Conveyor

- Designed an automated conveyor system integrating sensors, OpenCV-based computer vision, and embedded control.
- Implemented PI motor speed control using encoder feedback for closed-loop operation.

RTA AI Intelligent Traffic Management Solution

- Developed an AI-powered smart traffic management concept addressing congestion reduction and emergency vehicle prioritization.
- Presented the solution including business strategy, Responsible AI framework, data strategy, and live prototype demonstration.

Water Efficient Solar Panel Cleaning Mechanism

- Designed an automated solar panel cleaning system achieving 98.1% water reduction through optimized mechanical design.
- Performed stress, fatigue, buckling, and deflection analysis to validate structural performance.

Electro-Pneumatic & Hydraulic Control System

- Designed pneumatic circuits and programmed control logic for automated double-acting cylinder operation.
- Developed a suction-based glass handling mechanism using industrial automation principles.

Smart Street Lighting System

- Developed an adaptive lighting system using Raspberry Pi, LDR, and motion sensors achieving simulated energy savings.

Certifications & Professional Development

- RTA Innovation Challenge Certificate - AI, Responsible AI, Data Strategy, and Innovation
- NVIDIA - Generative AI Explained
- Cisco Networking Academy - Introduction to Modern AI
- Cisco Networking Academy - Apply AI: Analyze Customer Reviews
- Cisco Networking Academy - Introduction to Internet of Things (IoT)
- Arduino UNO Q Robocar Workshop - Besomi Academy
- Certificate of Appreciation - Canadian University Dubai

Technical Skills

Programming: Python, C/C++, MATLAB, Arduino Programming, Structured Text (PLC)

Artificial Intelligence & Computer Vision: Machine Learning, Generative AI, Object Detection, OpenCV, Image Processing, Mistral AI

Embedded Systems & Electronics: Arduino, Raspberry Pi, Arduino UNO Q, Sensors, Ultrasonic Sensors, LDR Sensors, CAN Bus, ECU Diagnostics

Control Systems & Automation: PI Control, Feedback Control Systems, PLC Logic, Pneumatics, Hydraulics, Process Control

Engineering Software: AutoCAD, MATLAB Simulink, Connected Components Workbench (CCW), BMW ISTA

Mechanical Engineering: 3D CAD Modelling, Mechanical Design, Stress Analysis, Fatigue Analysis, Failure Analysis, Manufacturing Processes